AnalysisTools user manual

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1. Introdution

AnalysisTools is a set of utilities to (mainly) analyse trajectories of particle-based simulations. The 34 utilities (including one from the DL_MESO simulation package can be roughly divided into four types:

- utilities to calculate system-wide properties; e.g., pair correlation functions or particle densities along a simulation box axis
- utilities to calculate per-molecule or per-aggregate properties (where aggregate stands for any supramolecular structure); e.g., shape descriptors for individual molecules or whole micelles
- utilities to manipulate a configuration; e.g., create initial configuration for a simulation from scratch or by adding molecules to an existing one
- helper utilities to analyse text files; e.g., calculate averages and standard deviations of a data series

The `Examples` directory contains examples showcasing capabilities of individual utilities as well as possible workflows chaining several utilities.

1.1 Installation

AnalysisTools requires `C` and `Fortran` compilers and `cmake`. The utilities should be compiled in a separate directory, typically `<root>/build` (where `<root>` is the AnalysisTools root directory). The following generates a `Makefile` within `<root>/build`:

```
mkdir <root>/build; cd <root>/build; cmake ../
```

To then compile the utilities, simply run `make` (to compile all utilities) or `make <utilitiy name>` (to compile a single utility) in `<root>/build`. The binaries are located in the `<root>/build/bin` directory.

2. Format of input/output files

This section describes several file types used by many of the AnalysisTools utilities. Output files for the utilities themselves are described in their respective sections in Chapter 3.

Input files are divided into two categories: structure and coordinate files. The structure files contain information about the system composition, that is, number of beads and molecules, bead properties (mass, charge, etc.), molecule properties (number and types of beads, connectivity, etc.). The coordinate files contain individual timesteps with beads' coordinates (and, possibly, velocities, forces, etc.). Some files can be used as both a structure and a coordinate file, but using a different structure file may be essential. For example, an `xyz` file (see below) may be used as both, but because it only provides bead names, using a separate structure file would provide additional information (bead and/or molecule properties).

For working with aggregates, AnalysisTools' `agg` file format (see below) describes their compositions in terms of molecule numbers from a structure file used to generate it. Any of the `Aggregates` (TODO: reference) utilities can create the `agg` file.

2.1 Structure files

As structure files, any of the following formats can be used; example files for a simple 'sandbox' system are provided in the `Examples/InputFiles` directory. Utilities load a specific structure using the `-i <file>` option; if the option is missing, it is usually assumed the structure file is the same as the coordinate file.

`Info` utility (TODO: reference) can convert these formats between each other via the `-o <file>` option.

2.1.1 VTF format

See here for complete description and `in.vtf` and `in.vsf` for example files (a `vtf` file contains a structure part followed by a coordinate part, but a `vsf` file contains only the structure part). In this format, every bead is defined on its own line and all bonds all listed.

Not all keywords listed on the web page for the 'Atom lines' are recognized; keyword `n[ame]` is mandatory and optional keywords are `m[ass]`, `charge|q`, `r[adius]`,

resid, and res[name]. For the <aid-specifier>, only a single bead index number or the default keyword can be used. Other keywords are ignored.

AnalysisTools groups beads (and molecules) into bead (and molecule) types. Generally, only name defines types of beads, i.e., should two beads share the name, they will be of the same type; their mass, charge, and radius will each equal to that for the bead type's topmost atom line containing the corresponding keyword. If the keyword is missing from all atom lines, that characteristic is undefined. The --detailed command line option triggers detailed recognition, where beads that share the name but differ in mass, charge, and/or radius will be of different type.

Different molecule types, on the other hand, are always distinguished based on all their characteristics, i.e., name, connectivity, and order of beads (order of its vsf indices). However, molecule types can also be affect by the --detailed option because of the bead type definition

For the effect of the --detailed option, see the Examples/Info directory.

For bond lines, only single bonds are recognized (i.e., bond <index1>:<index2> format).

Periodic boundary condition can be included as pbc <x> <y> <z> for a cuboid box or pbc <a> <b> <c> <alpha> <beta> <gamma> for a general paralleloid box.

2.1.2 FIELD format

A format similar to the FIELD input file for DL_MESO_DPD simulation package; see in.FIELD for an example. This concise format defines species of beads and molecules, providing number of beads/molecules of given species as well as their properties.

The species section is the same as in the DL_MESO FIELD file, but the molecules section differs slightly. Besides the beads part, every molecule can have bonds, angles, and dihedrals, but even though up to three parameters for potentials are read, AnalysisTools does not recognize the potential keywords (harm, cos, etc.). These potential parameters are only used for generating new structure files via the Info utility (TODO: reference). The molecules section can also contain one extra part: impropers for improper angles which has the same format as the dihedrals section (this was added because the LAMMPS molecular dynamics simulator distinguishes between the two).

Note that the FIELD file does not have to contain Interactions section; this part is ignored.

To be recognized by AnalysisTools utilities, the file must either be called FIELD or have a .FIELD extension.

2.1.3 XYZ format

This is the simplest and probably best known format, see, e.g., here for the description and `in.xyz` for an example. An `xyz` file is essentially a file with timesteps (bead coordinates), and the only structure information are bead names.

The comment line at the beginning of each timestep can contain periodic boundary conditions as `<x> <y> <z>` (cuboid box) or `<a> <b> <c> <alpha> <beta> <gamma>` (general paralleloid box); it is similar to the `pbc` keyword in `vtf` file. To use this box definition, use `-pbc <n>` option where `<n>` is the position of the first number on the comment line (e.g., `1` if the comment line starts with the box size).

2.1.4 LAMMPS data format

This rather complicated format comes from the LAMMPS molecular dynamics simulator and lists the numbers of beads, bonds, angles, etc., as well as individual beads, bonds, angles, etc., and information about potentials for bonds, angles, etc. See here for description of the format and `in.data` for an example.

AnalysisTools reads the header information as well as `Masses`, `Atoms`, `Velocities`, `Bonds`, `Angles`, `Dihedrals`, `Impropers`. It also reads `Bond Coeffs`, `Angle Coeffs`, `Dihedral Coeffs`, and `Improper Coeffs` sections.

The `Atoms` section must have `atom_style full` format, that is every bead line must have the format `<id> <mol id> <bead type> <q> <x> <y> <z>`, where `<id>` is a bead index, `<mol id>` is a molecule index (-1 means the bead is in no molecule), `<bead type>` is the bead type index corresponding to a line in the `Masses` section, `<q>` is charge, and `<x> <y> <z>` are coordinates.

For the `Bond Coeffs` and `Angle Coeffs` sections, harmonic potential is assumed, and the first value of each bond/angle type is multiplied by 2 because LAMMPS uses harmonic spring strength of $k/2$ (as opposed to, e.g., DL_MESO which uses k).

For the `Dihedral Coeffs` and `Improper Coeffs` sections, AnalysisTools reads at most three numbers without assuming any specific potential.

All the `Coeffs` sections are used only for generating new structure files via the `Info` utility, TODO: reference (specifically, only to generate `FIELD` and LAMMPS `data` files because no other supported file contains this information).

To be recognized by AnalysisTools utilities, the file must have an extension `.data`.

2.1.5 LAMMPS lammpstrj format

This mainly coordinate file format comes from the LAMMPS dump style custom command. See `in.lammpstrj` for an example.

Of the possible LAMMPS attributes, AnalysisTools recognizes only `id` (bead index – mandatory), `element` (bead name), `x y z` (bead coordinates), `vx vy vz` (bead velocities), and `fx fy fz` (bead forces). Any other attribute is ignored. Note that if `element` is missing, all beads are considered of the same type called `bt`.

To be recognized by AnalysisTools utilities, the file must have an extension `.lammpstrj`.

2.2 Coordinate files

As coordinate files, any of the following formats can be used; example files for a simple 'sandbox' system are provided in the `Examples/InputFiles` directory. These files can be combined with any of the above-described structure files (via the `-i <file>` option), but some may be used as both the structure and coordinate file.

`SelectedVcf` utility (TODO: reference) can generate any of these file types, converting between the formats; all utilities that require coordinate file can use any of these formats.

2.2.1 VTF format

See here for the description of a coordinate part of the `vtf` format and `in.vtf` and `in.vcf` for example files (a `vtf` file contains the coordinate part after a structure part, and a `vcf` file contains only the coordinate part).

AnalysisTools recognizes both ordered and indexed timesteps as well as a `pbc` line defining box dimensions.

If an AnalysisTools utility is not provided with an extra structure file (`-i <file>` option), the same filename is assumed but with the `.vsf` extension instead of the `.vcf` extension.

2.2.2 XYZ format

This is the simplest and probably best known format, see, e.g., here for the description and `in.xyz` for an example.

The comment line at the beginning of each timestep can contain periodic boundary conditions as `<x> <y> <z>` (cuboid box) or `<a> <b> <c> <alpha> <beta> <gamma>` (general paralleloid box); it is similar to the `pbc` keyword in `vtf` file. To use this box definition, use `-pbc <n>` option where `<n>` is the position of the first number on the comment line (e.g., `1` if the comment line starts with the box size).

The `xyz` coordinate file will be used as a structure file as well unless an extra structure file is provided via the `-i <file>` option.

2.2.3 LAMMPS lammpstrj format

This coordinate file format comes from the LAMMPS dump style custom command. See `in.lammpstrj` for an example.

Of the possible LAMMPS attributes, AnalysisTools recognizes only `id` (bead index – mandatory), `element` (bead name), `x y z` (bead coordinates), `vx vy vz` (bead velocities), and `fx fy fz` (bead forces). Any other attribute is ignored. Note that if `element` is missing, all beads are considered of the same type called `bt`.

To be recognized by AnalysisTools utilities, the file must have an extension `.lammpstrj`.

This coordinate file will be used as a structure file as well unless an extra structure file is provided via the `-i <file>` option.

2.3 Aggregate file

TODO: rewrite

A `file.agg` is generated using `Aggregates` (or `Aggregates-NotSameBeads`) utility. The file contains information about the number of aggregates in each timestep and which molecules and monomeric (i.e., unbonded) beads belong to which aggregate. It serves as an additional input file for utilities that calculate aggregate properties; `agg` file is, therefore, linked to the `vcf` file that was used to generate it.

The `agg` file is a simple text file. The first line contains the command used to generate it – parts of this command may be necessary for subsequent analysis of aggregates. The second line is blank, and from the third line, the data for individual timesteps are shown. It follows these rules:

- each timestep starts with `Step: <int>` (only `Step` keyword is read by the utilities)
- the second line is the number of aggregates in the given timestep and is followed by a blank line
- there are two lines for each aggregate:
  1) number of molecules in the aggregate followed by their indices taken from the `vsf` file (`resid` indices)
  2) number of monomeric beads in the aggregate followed by their indices taken from the `vsf` file (`atom` indices)
- no blank or comment lines are allowed inside the aggregate block
- not all molecules present in the `vcf` file used to generate this file must be present in every timestep

Note that the term aggregate also refers to free chains (i.e., fully dissolved chains).

When the keyword `Last` is present instead of `Step`, it signalises the end of the `agg` file; no utility will read anything beyond this keyword.

Besides using this file for further analysis by other utilities, the indices can be used in vmd to visualize, e.g., only a specific aggregate by using `resid 1 2 3` in the `Selected Atoms` box in vmd.

An example of an `agg` file can be found in the `Examples/DistrAgg` directory.

3. Utilities

All utilities have command line options affecting their behaviour. Most of the utilities share several standard command line options.

The following options are for all utilities and are not described with the utilities themselves.

General options common to all utilities	
`--verbose`	print information about all bead and molecule types to the screen
`--silent`	run silently, i.e., without any output at all (overrides `--verbose` option)
`--help`	print short description of the utility and usage
`--version`	print version of the utilities and exit

The following options are used by most utilities, and because they affect their behaviour, they are also listed in the utilities' descriptions.

Options for structure and coordinate input files	
`-i <name>`	use specified structure file, see Section 2.1 for possible formats
`--detailed`	only `vtf` structure file: detailed recognition for bead (and, consequently, molecule) types based on the bead names as well as charges, masses, radii
`--variable`	only `vtf` coordinate file: assume indexed timesteps with variable numbers of beads (otherwise, the same number of beads in each timestep is required)
`-pbc <n>`	only `xyz` file: position of the first box dimension in the file's comment line at the beginning of each timestep
`-st <n>`	any coordinate file: start calculations at `n`-th timestep
`-e <n>`	any coordinate file: end calculations at `n`-th timestep
`-sk <n>`	any coordinate file: skip every `n` timesteps during calculations

For `vtf` structure files, the bead types are defines solely based on their name; the other properties (mass, charge, and radius) are taken from the first `a[tom]` line with the bead of that name that contains the appropriate keyword. For example,

lines

```
atom default name A m 1
atom 1 name B
atom 2 name B m 2
atom 3 name A m 2
atom 4 name A
```

would define two bead type called `A` and `B` with masses equal to 1 and 2, respectively. Using the `--detailed` option would split the `A` beads into three types: `A` and `A_1` with masses equal to 1 and 2, respectively, and `A_2` with undefined mass (it remains undefined because there is an ambiguity—should the last bead have mass 1 or 2?). On the other hand, the `B` beads would be assigned the same bead type with mass equal to 2 (there is no ambiguity because only one mass is specified for the `B` beads). See `Examples/Info` directory for a more complete example.

For `vtf` coordinate file with indexed timesteps, all timesteps are expected to have the same number of beads (though not necessarily the same beads). If some timesteps contain different number of beads, `--variable` option must be used. The option is implemented because without knowing in advance the number of beads in a timestep, reading that timestep is slower (significantly so for large systems).

Plain `xyz` format provides only bead names and coordinates, but the comment line at the beginning of every timestep can store extra information. Here, only box dimensions are read, but because there are extended `xyz` formats containing more information, the `-pbc` option is used to specify where the box dimensions are (as well as to tell the utility there are box dimensions at all). For example, `-pbc 3` must be specified for the following comment line

`something more 10 10 10 90 60 30`.

This would define a paralleloid box with $a = b = c = 10$ and angles $\alpha = 90°$, $\beta = 60°$, and $\gamma = 30°$. If less than 6 numbers were provided, only the first three would be used to define a cuboid box.

3.1 Average

This utility calculates one of three types of averages based on a used option from specified column(s) of data in the input text file (all `#`-starting lines and blank lines are ignored). It either produces an overall average with statistical error and an autocorrelation time estimate (`-tau` option), block averages (`-b` option), or moving averages (`-m` option). While the `-tau` option appends a single line to the output file, either of the `-b` or `-m` options creates a new output file with somewhat smoother data.

The first and last lines used for the average calculation can be controlled using the standard `-st` and `-e` options.

3.1.1 Estimate autocorrelation time via -tau option

The average value of an observable $\mathcal{O}$ is a simple arithmetic mean,

$$\langle \mathcal{O} \rangle = \frac{1}{N} \sum_{i=1}^{N} \mathcal{O}_i, \tag{3.1}$$

where N is the number of measurements and the subscript i denotes individual measurements. If the measurements are independent (i.e., uncorrelated), the statistical error, ϵ, is given by:

$$\epsilon^2 = \frac{\sigma^2_{\mathcal{O}_i}}{N}, \tag{3.2}$$

where $\sigma^2_{\mathcal{O}_i}$ is the variance of the individual measurements,

$$\sigma^2_{\mathcal{O}_i} = \frac{1}{N-1} \sum_{i=1}^{N} (\mathcal{O}_i - \langle \mathcal{O} \rangle)^2. \tag{3.3}$$

For correlated data, the autocorrelation time, τ, representing the number of steps between two uncorrelated measurements must be determined. Every τ-th measurement is uncorrelated, so the equation (3.2) can then be used to estimate the error.

A commonly used method to estimate τ is the binning (or block) method. In this method, the correlated data are divided into N_B non-overlapping blocks of size k ($N = kN_\mathrm{B}$) with per-block averages, $\mathcal{O}_{\mathrm{B},n}$, defined as:

$$\mathcal{O}_{\mathrm{B},n} = \frac{1}{k} \sum_{\substack{i=1+\\(n-1)k}}^{kn} \mathcal{O}_i. \tag{3.4}$$

If $k \gg \tau$, the blocks are assumed to be uncorrelated and equation (3.2) can be used:

$$\epsilon^2 = \frac{\sigma^2_\mathrm{B}}{N_\mathrm{B}} = \frac{1}{N_\mathrm{B}(N_\mathrm{B}-1)} \sum_{n=1}^{N_\mathrm{B}} (\mathcal{O}_{\mathrm{B},n} - \overline{\mathcal{O}})^2. \tag{3.5}$$

An estimate of the autocorrelation time can be obtained using the following formula:

$$\tau_\mathcal{O} = \frac{k\sigma^2_\mathrm{B}}{2\sigma^2_{\mathcal{O}_i}}. \tag{3.6}$$

The number of blocks, N_B, is supplied as an argument of the `-tau` option and Average then appends a single line to the `<output>` file; the line starts with N_B and

continues with three values ($\langle\mathcal{O}\rangle$, ϵ, and $\tau_{\mathcal{O}}$) per every data column specified in the Average command.

A way to quickly get a τ estimate is to use a wide range of N_{B} values and plot $\tau_{\mathcal{O}}$ from equation (3.6) as a function of N_{B}. Because the number of data points in one block should be significantly larger than the autocorrelation time (e.g., ten times larger), plotting $f(x) = N/(10x)$ will produce a monotonously decreasing curve that intersects the $\tau_{\mathcal{O}}$ vs. N_{B} curve. A value of $\tau_{\mathcal{O}}$ near the intersection (but to the left, where the decreasing curve is above $\tau_{\mathcal{O}}$ vs. N_{B} curve) can be considered a safe estimate of τ.

3.1.2 Block averages via -b option

Besides estimating the autocorrelation time, the per-block averages can be themselves plotted to get (probably) smoother dataset. Using the `-b` option, the number of datapoints per block to average, k, is supplied, and the utility prints the per-block averages from equation (3.4) to the output file.

This way of averaging could be useful for example with density data produced by DensityBox or related utilities; if the bin width supplied to the DensityBox was too small, it is (possibly much) faster to block-average the densities rather than rerun DensityBox. For this case, specify the first column (distance) along with any density columns from the density file.

3.1.3 Moving averages via -m option

More common way to smoothing noisy data is to use the moving (or rolling or running) average (or moving mean or rolling mean); this common method does have many names.

Similarly to the block-average, the first element of the moving average is a simple mean of k values. Unlike with block-average, however, the k values for the next element are obtained by ignoring only one value and taking the next k values; i.e., $k-1$ values from the previous element of the moving average are always reused as opposed to the block-average where the blocks of data are not overlapping.

The moving average elements, $\mathcal{O}_{\mathrm{M},n}$, are defined as

$$\mathcal{O}_{\mathrm{M},n} = \frac{1}{k}\sum_{n}^{n+k-1} \mathcal{O}_i. \tag{3.7}$$

Usage:

```
Average <input> <output> <column(s)> <options>
```

Mandatory arguments	
`<input>`	input text file
`<output>`	output text file
`<column(s)>`	at least one column number from `<input>`

Options	
`-tau <int>`	τ estimation where `<int>` represents the number of blocks N_B
`-b <int>`	block-average printing mode where `<int>` represents the number of datapoints per one block, k (equation (3.4))
`-m <int>`	moving average mode where `<int>` represents the number of datapoints per one moving block, k (equation (3.7))
Other options (see the beginning of Chapter 3)	
`-st`, `-e`, `--verbose`, `--silent`, `--help`, `--version`	

3.2 DensityBox

This utility calculates number density for all bead types along all three axes directions of the simulation box. The density is calculated from 0 to box length in the given direction, that is, the box is 'sliced' into blocks with width `<width>` and numbers of different bead types are counted in each of the slices. Or to put it more rigorously, a density profile for a bead type i along an axis α, $\rho(\alpha)$, is calculated as

$$\rho_i(\alpha) = \frac{\sum_j \delta(\alpha - \alpha_j^i)}{\Delta\alpha L_\beta L_\gamma}, \tag{3.8}$$

where $\delta(\alpha - \alpha_i^j)$ gives the number of i beads inside a slice of the simulation box of the thickness $\Delta\alpha$ along the α-axis; L_β and L_γ represent box side lengths along the two remaining axes.

The utility does not distinguish between beads with the same name in different molecules, so if one bead type is in more than one molecule type, its density will be averaged over all molecule types it appears in. However, the `-x` option can be used to first run the utility without one molecule type and then rerun it, excluding the other molecule type. Thus, two output files (per axis) are generated, each missing densities from the specified molecule types.

Note that this utility assumes orthogonal box with constant side lengths; in case of triclinic box and/or varying box size, undefined behaviour may occur (i.e., the utility crashes), and any results will not be reliable.

Also note that the same density profile (along with other properties) can also be calculated via the Local utility (3.4).

Usage:

```
DensityBox <input> <width> <output> <options>
```

Mandatory arguments	
`<input>`	input coordinate file (either `vcf` or `vtf` format)
`<width>`	width of each bin of the distribution
`<output>`	three output files with automatic `-x.rho`, `-y.rho`, and `-x.rho` endings
Non-standard options	
`-x <mol name(s)>`	exclude specified molecule type(s) (i.e., do not calculate density for beads in molecules `<mol name(s)>`)
Other options (see the beginning of Chapter 3)	
`-st`, `-e`, `-sk`, `-i`, `--variable`, `-pbc`, `--detailed`, `--verbose`, `--silent`, `--help`, `--version`	

Format of output files:
1) `<output>` – bead densities
- first line: command used to generate the file
- second line: column headers
  - first is the centre of each bin (governed by `<width>`); i.e., if `<width>` is 0.1, then the centre of bin 0 to 0.1 is 0.05, centre of bin 0.1 to 0.2 is 0.15, etc.
  - the rest are for the calculated data: each column corresponds to the number density of the specified bead type

3.3 Info

This utility prints the system information read from an input structure file, possibly augmenting it with information from another structure file (`-i[!] <file>` option) and/or pruning the system to contain only beads from a provided coordinate file (`-c` option). For the effects of these options, see the `Examples/Info` folder.

`Info` can also write this information into a new structure file of the given format (`-o` option). Allowed output files are `vtf` structure file (with the extension `.vsf`), LAMMPS `data` file (with the extension `.data`), and DL_MESO FIELD file (called `FIELD` or with the extension `.FIELD`). Coordinates (and, possibly, velocities) from the coordinate file are also used for output LAMMPS `data` file.

Usage:

```
Info <input> <options>
```

Mandatory argument	
`<input>`	input structure file
Options for input files	
`-i[!] <file>`	extra input structure file; see `Examples/Info` directory for the effect of `-i` and `-i!` options (default: None)
`-c <file>`	input coordinate file (default: None)
`--detailed`	only `vtf` structure file: specify recognition of bead (and molecule types) according to more than the bead name
`-pbc <n>`	only `xyz` file: specify position of box dimensions on a comment line (default: no box dimensions)
`--variable`	only `vtf` coordinate file: specify indexed timesteps with variable number of beads (only affects the results if `-st` is used)
`-st <n>`	any coordinate file: use `n`-th timestep
Options for output file	
`-o <file>`	output coordinate file (default: None)
`-def <bead>`	only `vtf` structure file: use bead type `<bead>` for the 'atom default' line (default:bead type with the most unbonded beads)
`--mass`	only `data` structure file: define LAMMPS atom types by mass but print per-atom charges in the `Atoms` section (default: atom types are the same as `Info`-recognized bead types)
Other options (see the beginning of Chapter 3)	
`-st`, `--variable`, `-pbc`, `--detailed`, `--verbose`, `--silent`, `--help`, `--version`	

3.4 Local

This utility calculates various properties both locally (i.e., average profiles along the three principal axes) and globally (i.e., per-timestep averages from the whole system), producing four output files (one file per axis with profiles along that axis and one file for global averages).

Primarily, the utility calculates kinetice energy and temperature (which requires input coordinate file with velocities) and density profiles, but using the `--pot` switch, pair-wise potential can be included to calculate pressure. For now, only shifted Sutton-Chen potential is implemented (see below).

Unlike the DensityBox utility (Section 3.2), Local takes into account changing size of an orthogonal simulation box, calculating the profiles along the three axis with respect to the geometric centre of all beads (or the centre of mass if `--com` is

used); the resulting profiles (averaged over all specified timesteps) are printed with 0 at the average geometric centre (or the centre of mass).

An extra file can be provided via the `-fx <file>` option. This file defines the inter-bead potential to calculate and may contain quantities for reduced units or coarse-graining level. The section of potential parameters must start with the line `potential <potential>` and end with the `finish` keyword. Information for every bead type (named according to the used structure file) should be present in this section (extra data for bead types not present in the system are ignored). The following is full structure of the `potential` section:

```
potential <potential>       # potential type to use
<global parameter 1>        # parameter(s) identical to all beads
...
<global parameter N>
<bead>                      # name of the first bead type
<parameter 1> <value(s)>    # first parameter of <potential> for bead type
...                            <bead> (see below for parameter description)
<parameter 2> <value(s)>    # last parameter for <potential> and <bead>
<bead>                      # name of the next bead type
<pot_param_1> <value(s)>
...                         # parameters for the second bead type
<pot_param_N> <value(s)>
finish                      # end of the potential section
```

Besides the potential, self-explanatory lines with keyword-value pairs may specify reduced units—`energy <value>`, `temperature <value>`, `length <value>`, `mass <value>`, and `pressure <value>`. Provided potential parameters are divided by these values to be in reduced units while results are multiplied by these values to provide data in real units; input coordinates and velocities are assumed to be in reduced units.

Comments (#-initiated lines) and blank lines in the file are ignored.

3.4.1 Sutton-Chen (SC) and shifted SC potentials

TODO: implement vanilla Sutton-Chen...

This original density-dependent potential based on embedded atom model was developed to study metals[1] and is defined as

$$U = u^{\text{rep}} + u^{\text{dd}} = \epsilon \sum_i \sum_{j>i} \left(\frac{a}{r_{ij}}\right)^m - c\epsilon \sum_i \sqrt{\rho_i}, \text{ where } \rho_i = \sum_{j\neq i} \left(\frac{a}{r_{ij}}\right)^n, \tag{3.9}$$

where ϵ is an energy parameter, a is the lattice constant, c is dimensionless parameter, m and n are integer parameters ($m > n$), ρ_i is local density for bead i, r_{ij} is the distance between beads i and j. The pair-wise repulsive part representing Van der Waals repulsion gives rise to force

$$f_{ij}^{\text{rep}} = -\frac{\mathrm{d}u^{\text{rep}}}{\mathrm{d}r_{ij}} = \frac{\epsilon_{ij} m_{ij}}{r_{ij}} \left(\frac{a_{ij}}{r_{ij}}\right)^{m_{ij}} \tag{3.10}$$

and the density-dependent part representing metal bonds to force

$$f_{ij}^{\text{dd}} = -\frac{n_{ij}}{2} \left(\frac{c_i \epsilon_{ii}}{\sqrt{\rho_i}} + \frac{c_j \epsilon_{jj}}{\sqrt{\rho_j}}\right) \left(\frac{a_{ij}}{r_{ij}}\right)^{n_{ij}}. \tag{3.11}$$

To get potential that is smooth at the cut-off distance, r_{c}, a shifted variant can be used:

$$U^{\text{sf}} = u(r) - u(r_{\text{c}}) - \left.\frac{\mathrm{d}u}{\mathrm{d}r}\right|_{r_{\text{c}}} (r - r_{\text{c}}), \tag{3.12}$$

where $u(r)$ is the standard SC in Equation(3.9). The repulsive part is

$$u^{\text{sf-rep}} = \epsilon_{ij} \left(\frac{a_{ij}}{r_{ij}}\right)^{m_{ij}} - \left(\frac{a_{ij}}{r_{\text{c}}}\right)^{m_{ij}} + m_{ij} \left(\frac{a_{ij}}{r_{\text{c}}}\right)^{m_{ij}} \frac{r_{ij} - r_{\text{c}}}{r_{\text{c}}} \tag{3.13}$$

and the local density, ρ_{ij}, for the density-dependent part, $u^{\text{sf-dd}} = c\epsilon \sum_i \sqrt{\sum_{j \neq i} \rho_{ij}}$, is

$$\rho_{ij} = \left(\frac{a_{ij}}{r_{ij}}\right)^{n_{ij}} - \left(\frac{a_{ij}}{r_{\text{c}}}\right)^{n_{ij}} + n_{ij} \left(\frac{a_{ij}}{r_{\text{c}}}\right)^{n_{ij}} \frac{r_{ij} - r_{\text{c}}}{r_{\text{c}}} \tag{3.14}$$

The repulsive and density-dependent forces can then be written as

$$f_{ij}^{\text{sf-rep}} = \frac{\epsilon_{ij} m_{ij}}{r_{ij}} \left(\frac{a_{ij}}{r_{ij}}\right)^{m_{ij}} + m_{ij} \epsilon_{ij} \left(\frac{a_{ij}}{r_{\text{c}}}\right)^{m_{ij}} \frac{r_{ij}}{r_{\text{c}}} \tag{3.15}$$

and

$$f_{ij}^{\text{sf-dd}} = \frac{1}{2} \left(\frac{c_i \epsilon_{ii}}{\sqrt{\rho_i}} + \frac{c_j \epsilon_{jj}}{\sqrt{\rho_j}}\right) \left[n_{ij} \left(\frac{a_{ij}}{r_{ij}}\right)^{n_{ij}} - n_{ij} \left(\frac{a_{ij}}{r_{\text{c}}}\right)^{n_{ij}} \frac{r_{ij}}{r_{\text{c}}}\right] \tag{3.16}$$

The cross-terms for different bead types are

$$m_{ij} = \frac{m_{ii} + m_{jj}}{2},\ n_{ij} = \frac{n_{ii} + n_{jj}}{2},\ \text{and}\ \epsilon_{ij} = \sqrt{\epsilon ii \epsilon_{ij}}. \tag{3.17}$$

Finally, the quasi-coarse-grained dynamics scheme by Dongare[2] is used, where the coarse-graining level (CG) represents the number of crystal unit cells that are reduced into one in each direction; i.e., CG level i means $i \times i \times i$ unit cells are represented by a single CG unit cell. The potential parameters are then coarse-grained

via the following equations, where subscript CG represents the coarse-grained parameter and i the CG level,

$$\begin{aligned} n_{\mathrm{CG}} &= n \\ m_{\mathrm{CG}} &= m \\ \epsilon_{\mathrm{CG}} &= i^3\epsilon \\ c_{\mathrm{CG}} &= c \\ T_{\mathrm{CG}} &= i^3 T \\ a_{\mathrm{CG}} &= ia \\ r_{\mathrm{c,CG}} &= ir_{\mathrm{c}} \end{aligned} \tag{3.18}$$

The following table shows the form of the potential section in the input file (`-fx <file>`).

Sutton-Chen potential (use `SuttonChen` as `<potential>`) Shifted Sutton-Chen potential (use `ShiftedSC` as `<potential>`)	
	2 global parameters:
`CG <int>`	coarse-graining level (see below)
`cutoff <float>`	cut-off distance, r_{c}
	4 lines per bead type:
`mn <int> <int>`	m and n parameters; lower number is assigned n
`c <float>`	dimensionless c parameter
`eps <float>`	energy parameter $\epsilon^\star$
`a <float>`	lattice constant, a

*if in real units, the value must be in Kelvins

References

[1] A. Sutton, J. Chen, *Philos Mag Lett* **1990**, *61*, 139–146.

[2] A. M. Dongare, *Philos Mag* **2014**, *94*, 3877–3897.

3.5 SelectedVcf

This utility creates a new coordinate file containing only beads of specified types; any of the `xyz` , `vcf` , or `lammpstrj` formats can be used as output.

Using `--reverse` switch, the bead types can also be specified in reverse, i.e., bead types to be excluded from the output file(s).

Which timesteps are saved can be controlled using several options: `-st` option for the first timestep to save and `-e` option for the last timestep to save; `-sk` for the number of ignored steps per one saved (i.e., `-sk 9` means every 10th timestep is saved). On the other hand, to explicitly specify which steps to save, the `-n` option is

used; SelectedVcf then ignores the `-st`, `-e`, and `-sk` options if present. A maximum of 100 steps can be specified for the `-n` option. Finally, using `--last` option saves only the last step from the input coordinate file (all the previous options are ignored if `--last` is used).

There is also an option to remove periodic boundary conditions for molecules (i.e., to join them) via `--join` switch. Conversely, the simulation box can be wrapped (i.e., the periodic boundary conditions applied, putting all beads inside the box) via `--wrap` switch. If both and `--wrap` and `--join` options are used, the simulation box is first wrapped and then the molecules are joined.

Usage:

```
SelectedVcf <input> <output> <bead type(s)> <options>
```

Mandatory arguments	
`<input>`	input coordinate file
`<output>`	output coordinate file
`<bead type(s)>`	bead type names to save (can be omitted if `--reverse` is used)

Options	
`--reverse`	reverse bead types to save, i.e., exclude `<bead type(s)>` instead of including them; if no `<bead type(s)>` are specified, all bead types are saved
`--join`	join molecules by removing periodic boundary conditions
`--wrap`	wrap simulation box (i.e., apply periodic boundary conditions)
`-n <int(s)>`	save only specified timesteps
`--last`	save only the last step

Other options (see the beginning of Chapter 3)
`-st`, `-e`, `-sk`, `-i`, `--variable`, `-pbc`, `--detailed`, `--verbose`, `--silent`, `--help`, `--version`